

Technical Report: Object-Centric Bimanual Coordinated Control

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Abstract

This report outlines the mathematical reformulation, and control architecture for a bimanual coordinated manipulation system within the `mc_rtc` framework.

1 System Architecture (Object-Centric Control)

Instead of a master and follower arm, connected by a virtual spring and damper, the system adopts a **object-centric approach**. A virtual object reference frame, denoted as $X_{0,\text{object}}$, is defined and dynamically tracked relative to the global world frame (0).

1.1 Code Framework

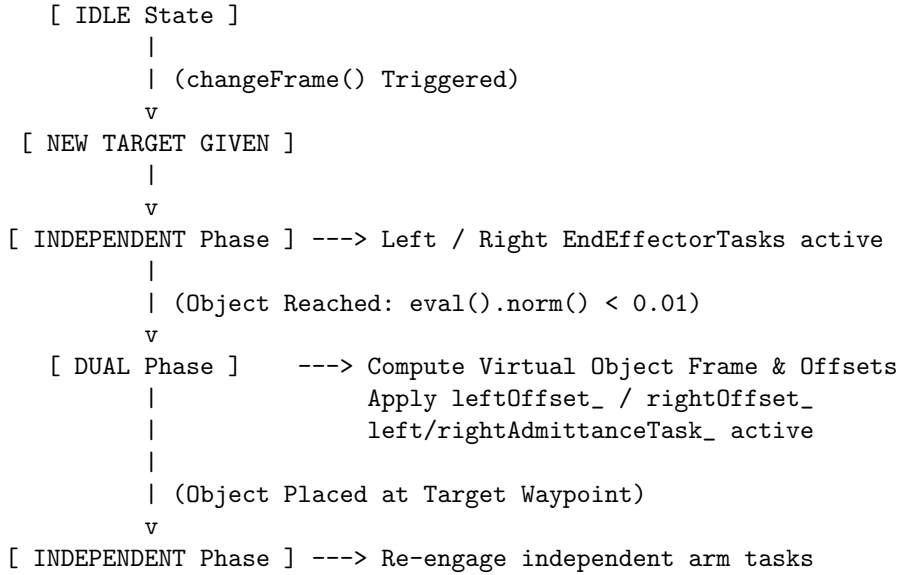


Figure 1: State machine pipeline for collaborative bimanual placement.

1.2 Object Frame Initialization

The initial virtual object pose is calculated based on the geometric midpoint between the left (L) and right (R) end-effectors.

$$p_{0,\text{object}} = \frac{p_{0,L} + p_{0,R}}{2} \quad (1)$$

$$R_{0,\text{object}} = R_{0,\text{base}} \quad (2)$$

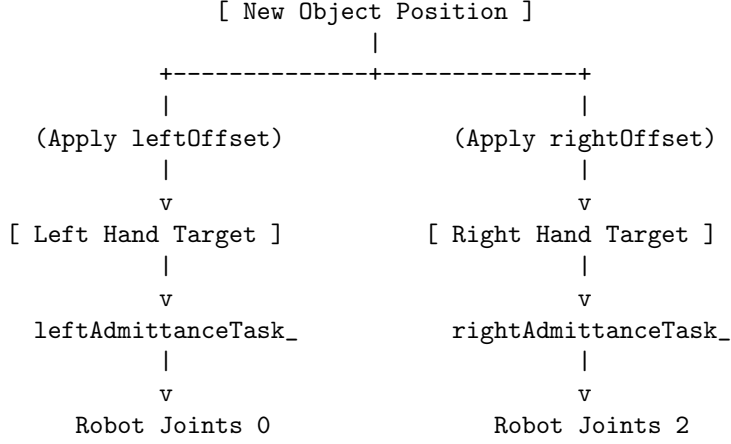


Figure 2: Data flow architecture of the dual-arm object tracking loop.

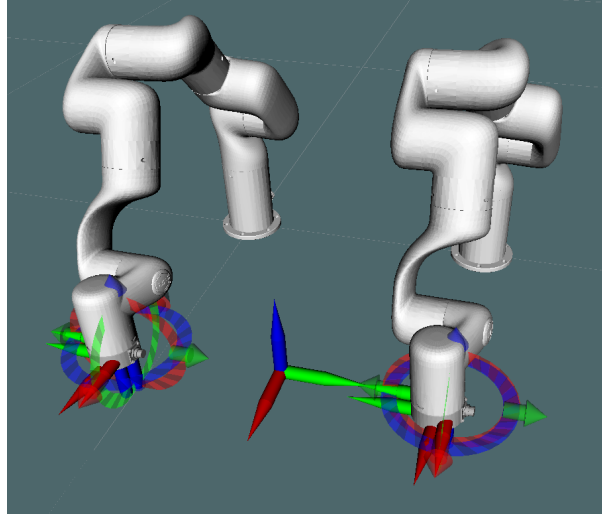


Figure 3: RVIZ simulation

2 Geometric Invariance via Relative Transformations

The grasp geometry is frozen at the activation time (t_0) by computing the relative transformations (offsets) of the end-effectors with respect to the virtual object frame:

$$X_{\text{object},L} = X_{0,L} \cdot X_{0,\text{object}}^{-1} \quad (3)$$

$$X_{\text{object},R} = X_{0,R} \cdot X_{0,\text{object}}^{-1} \quad (4)$$

These offsets ($X_{\text{object},L}$, $X_{\text{object},R}$) remain **strictly constant** throughout the entire manipulation task.

When the target object pose is commanded to follow a trajectory $X_{0,\text{object}}(t)$ over time, the global kinematic targets for the individual end-effectors are updated at each control loop.

$$X_{0,L}^{\text{target}}(t) = X_{\text{object},L} \cdot X_{0,\text{object}}(t) \quad (5)$$

$$X_{0,R}^{\text{target}}(t) = X_{\text{object},R} \cdot X_{0,\text{object}}(t) \quad (6)$$

3 Rationale and Key Advantages

1. **Stripped-Down Motion Planning Complexity:** By abstracting the bimanual system into a single virtual operational space.
2. **Perfect Geometric Cohesion:** Because the spatial transformations ($X_{\text{object},L}$, $X_{\text{object},R}$) are structurally invariant over time, both arms maintain an identical relative distance and orientation.
3. **Compliance and Tolerance Adaptation via Admittance:** To check the dual admittance tasks seamlessly absorb the error.